

Software Bill of Materials (CycloneDX)

Grounded in: BD Alaris™ Infusion System

Basis: CycloneDX 1.5; NTIA minimum elements.

Device	BD Alaris™ Infusion System (PCU software)
Format	CycloneDX 1.5 — machine + human readable; generated in CI
Note	Skeleton showing the expected structure/fields; component versions illustrative.

SBOM structure (CycloneDX)

```
{
  "bomFormat": "CycloneDX", "specVersion": "1.5",
  "metadata": {
    "component": { "type": "application", "name": "alaris-pcu-fw", "version": "" },
    "supplier": { "name": "Becton, Dickinson and Company" }
  },
  "components": [
    { "type": "library", "name": "openssl", "version": "3.0.x",
      "licenses": [{ "license": { "id": "Apache-2.0" } }],
      "properties": [{ "name": "support:level", "value": "supported" } ] },
    { "type": "library", "name": "network-stack", "version": "x.y" },
    { "type": "operating-system", "name": "rtos", "version": "x.y" }
  ],
  "vulnerabilities": []
}
```

What an excellent SBOM adds

- Supplier, license, version for every component (commercial, OSS, off-the-shelf).
- Support level and end-of-support date per component.
- A linked known-vulnerability list and a justification for any missing element.
- Continuous regeneration so the SBOM stays current through postmarket.

Sources (public)

- FDA / BD — “BD Receives FDA 510(k) Clearance for Updated BD Alaris Infusion System” (Jul 21, 2023).
- FDA — Cybersecurity in Medical Devices: QMS Considerations and Content of Premarket Submissions (final, Jun 27, 2025).
- AAMI SW96; ISO 14971; IEC 81001-5-1.